Appendix D – Complete SQL Script (.sql)

Read down the left column, then down the right. SQL content is retained as supplied; long lines wrap for display.

```
create database
centralized_and_transparent_grading_system
character set utf8mb4
collate utf8mb4_unicode_ci;

use centralized_and_transparent_grading_system;

create table Student(
student_id varchar(10) primary key,
full_name varchar(100) not null,
email varchar(100) not null unique,
password_hash varchar(255) not null,
program varchar(100) not null,
year_level tinyint unsigned not null,
section varchar(20) not null,
status enum('Active','Inactive') not null default 'Active',
created_at timestamp not null default
CURRENT_TIMESTAMP);

create table Teacher(
teacher_id varchar(10) primary key,
full_name varchar(100) not null,
email varchar(100) not null unique,
password_hash varchar(255) not null,
department varchar(50) not null,
employment_type enum('Full-time','Part-time') not null
default 'Full-time',
status enum('Active','Inactive') not null default 'Active',
created_at timestamp not null default
CURRENT_TIMESTAMP);

create table Admin(
admin_id int unsigned primary key auto_increment,
full_name varchar(100) not null,
email varchar(100) not null unique,
password_hash varchar(255) not null,
created_at timestamp not null default
CURRENT_TIMESTAMP);

create table Course(
course_code varchar(20) primary key,
course_name varchar(100) not null);

create table Class(
class_id int unsigned primary key auto_increment,
course_code varchar(20) not null,
section varchar(20) not null,
semester enum('1st Semester','2nd Semester') not null,
academic_year varchar(9) not null,
teacher_id varchar(10) not null,
constraint fk_class_course
foreign key(course_code) references Course(course_code)
on update cascade on delete restrict,
constraint fk_class_teacher
foreign key(teacher_id) references Teacher(teacher_id)
on update cascade on delete restrict,
unique key uq_class (course_code, section, semester,
academic_year));

create table Enrollment(
enrollment_id int unsigned primary key auto_increment,
student_id varchar(10) not null,
class_id int unsigned not null,
running_grade decimal(5,2) null,
enrolled_at timestamp not null default
CURRENT_TIMESTAMP,
constraint fk_enrollment_student
foreign key(student_id) references Student(student_id)
on update cascade on delete restrict,
constraint fk_enrollment_class
foreign key(class_id) references Class(class_id)
on update cascade on delete restrict,
unique key uq_enrollment(student_id, class_id));

create table Assessment(
assessment_id int unsigned primary key auto_increment,
class_id int unsigned not null,
term enum('Midterm','Finals') not null,
title varchar(100) not null,
assessment_type
enum('Quiz','Activity','Exam','Project','Attendance') not null,
total_score decimal(6,2) not null,
weight decimal(5,2) not null,
created_at timestamp not null default
CURRENT_TIMESTAMP,
constraint fk_assessment_class
foreign key(class_id) references Class(class_id)
on update cascade on delete cascade,
constraint chk_weight check(weight BETWEEN 0 AND 100),
constraint chk_total_score check(total_score > 0));

create table StudentGrade(
grade_id int unsigned primary key auto_increment,
enrollment_id int unsigned not null,
assessment_id int unsigned not null,
score decimal(6,2) null,
remarks varchar(255) null,
created_at timestamp not null default
CURRENT_TIMESTAMP,
updated_at timestamp not null default
CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
constraint fk_grade_enrollment
foreign key(enrollment_id) references
Enrollment(enrollment_id)
on update cascade on delete cascade,
constraint fk_grade_assessment
foreign key(assessment_id) references
Assessment(assessment_id)
on update cascade on delete cascade,
unique key uq_grade(enrollment_id, assessment_id));

create table RevisionRequest(
```

```
request_id int unsigned primary key auto_increment,
grade_id int unsigned not null,
requested_points decimal(5,2) null,
reason varchar(255) not null,
status enum('Pending','Under Review','Approved','Closed')
not null default 'Pending',
requested_at timestamp not null default
CURRENT_TIMESTAMP,
reviewed_at timestamp null,
reviewed_by varchar(10) null,
constraint fk_request_grade
foreign key(grade_id) references StudentGrade(grade_id)
on update cascade on delete cascade,
constraint fk_request_teacher
foreign key(reviewed_by) references Teacher(teacher_id)
on update cascade on delete set null);

create table AuditLog(
log_id bigint unsigned primary key auto_increment,
student_id varchar(10) null,
teacher_id varchar(10) null,
admin_id int unsigned null,
action varchar(100) not null,
entity_type varchar(50) not null,
entity_id varchar(20) not null,
old_value varchar(100) null,
new_value varchar(100) null,
logged_at timestamp not null default CURRENT_TIMESTAMP,
details varchar(500) null,
constraint fk_audit_student
foreign key(student_id) references Student(student_id)
on delete restrict,
constraint fk_audit_teacher
foreign key(teacher_id) references Teacher(teacher_id)
on delete restrict,
constraint fk_audit_admin
foreign key(admin_id) references Admin(admin_id)
on delete restrict,
constraint chk_audit_one_actor
check((student_id is not null) + (teacher_id is not null) +
(admin_id is not null) = 1),
index idx_entity (entity_type, entity_id));

(9,NULL,'P001',NULL,'Updated request
status','RevisionRequest','3','Pending','Approved','2026-10-11
07:30:00','Partial credit given for item 12'),
(10,NULL,'P002',NULL,'Updated request
status','RevisionRequest','4','Pending','Closed','2026-10-11
07:30:00','Answer did not match the key'),
(11,NULL,'P001',NULL,'Updated
score','StudentGrade','6','88/100','93/100','2026-10-11
07:30:00','Approved revision request: partial credit for item
12'),

(16,NULL,NULL,1,'Changed employment
type','Teacher','P003','Full-time','Part-time','2026-10-28
02:58:40',NULL),
(17,NULL,NULL,1,'Changed employment
type','Teacher','P003','Part-time','Full-time','2026-10-28
02:58:41',NULL),
(18,'25-00418',NULL,NULL,'Submitted revision
request','RevisionRequest','8',NULL,'Pending','2026-10-28
05:11:58','Activity 1: re-check');
```